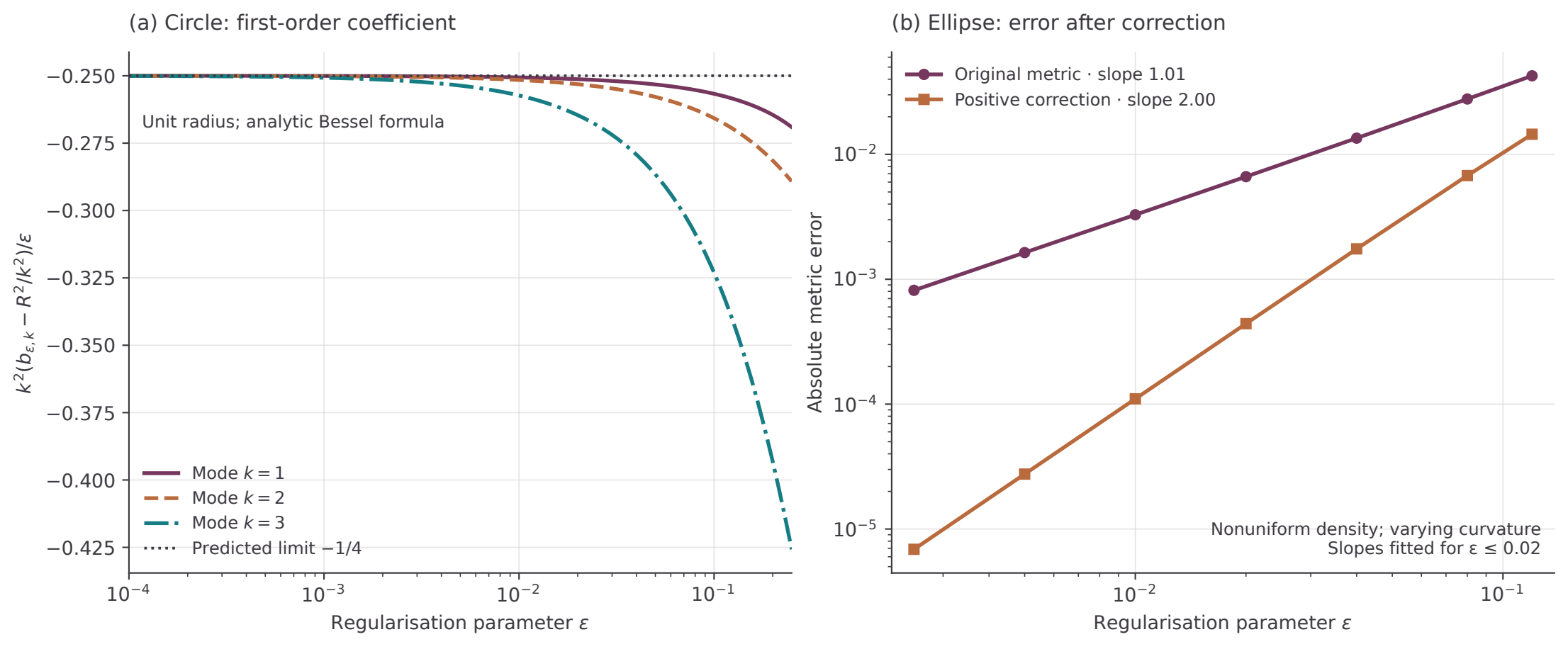
(a) Circle: first-order coefficient
Unit radius; analytic Bessel formula
Mode $k = 1$
Mode $k = 2$
Mode $k = 3$
Predicted limit −1/4
$k^2(b_{\varepsilon,k} - R^2/k^2)/\varepsilon$
−0.250
−0.275
−0.300
−0.325
−0.350
−0.375
−0.400
−0.425
10^{-4}
10^{-3}
10^{-2}
10^{-1}
Regularisation parameter ε
(b) Ellipse: error after correction
Original metric · slope 1.01
Positive correction · slope 2.00
Absolute metric error
10^{-2}
10^{-3}
10^{-4}
10^{-5}
Nonuniform density; varying curvature
Slopes fitted for ε ≤ 0.02
10^{-2}
10^{-1}
Regularisation parameter ε